

This file contains the English translation of the instructions used in the experiment. The original text is in Japanese. Note that all the screenshots provided below are from T1.

INSTRUCTIONS

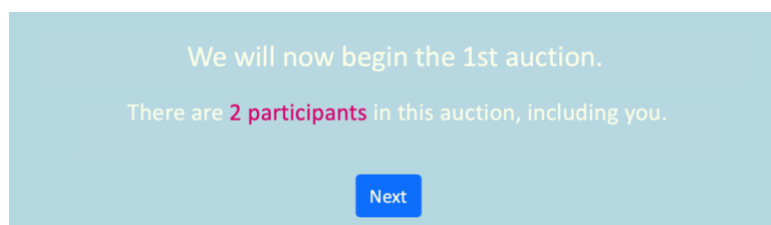
- Welcome to our experiment.
- You will participate in a series of auctions. The auctions will be repeated multiple times, but each auction will have different conditions, so please treat each auction as an independent event.
- In some rounds, you will participate in an auction with a group of 2 people, while in other rounds, the auction will involve a group of 4 people. The size of the group for each auction will change randomly.

[T1, T2, T3, T4]

- At the beginning of each round, the number of participants in the auction will be displayed as follows.

[T5, T6]

- The group size will be revealed during each auction, with further details explained later.



- In this experiment, you will be assigned one of four roles: A, B, C, or D. *You will keep the same role throughout the entire experiment.*
- For example, if you are assigned role A, you will remain in role A from the first auction until the last one.
- *The group members will change randomly for each round.*
- Please note that there will be multiple participants assigned to each role (A through D), so even if you end up in the same group as someone with the same role as before, that person may not be the same individual you were paired with previously.
- For instance, if you were paired with someone in role B during a round, even if you are in the same group as someone in role B again, it does not necessarily mean it will be the same person as before.

- A single item will be offered in each auction. The satisfaction you gain from winning the auction and acquiring the item is referred to as your "*evaluation value*."
- If you win the auction, you will receive a reward equivalent to this evaluation value. The evaluation value will be *different in each round*.
- It is randomly selected with *an equal probability (1/100) for each role*.
- Therefore, the evaluation value varies by role and can be any value between JPY 1 and JPY 1000, in increments of a JPY 10.
- You will know your own evaluation value, but you will not be able to know the evaluation values of others.

Buy-It-Now Stage:

- At the beginning of each auction, your evaluation value will be displayed along with a "Buy-It-Now Price."
- **This Buy-It-Now Price is set at ¥740 and will remain the same throughout the experiment.** Here is an example of the PC screen that will be displayed.

[T5, T6]

- You will not know the group size (2 or 4 members) until the Buy-It-Now stage is over.

Buy-It-Now Stage (Round 3, 4 Participants)

- Your evaluation value for this item is **790 yen**.
- If you proceed to the bidding stage without using the Buy-It-Now option, **your evaluation value will remain 790 yen**.
- The Buy-It-Now price for this item is **740 yen**.
- If you win, your profit will be calculated as follows:

Evaluation Value – Buy-It-Now Price
= 790 yen – 740 yen = 50 yen

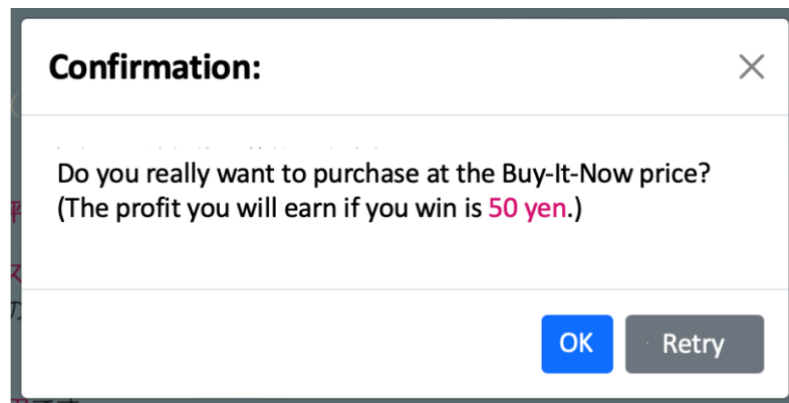
- Will you purchase at the Buy-It-Now price?
 (Time remaining: 0:22)

YES

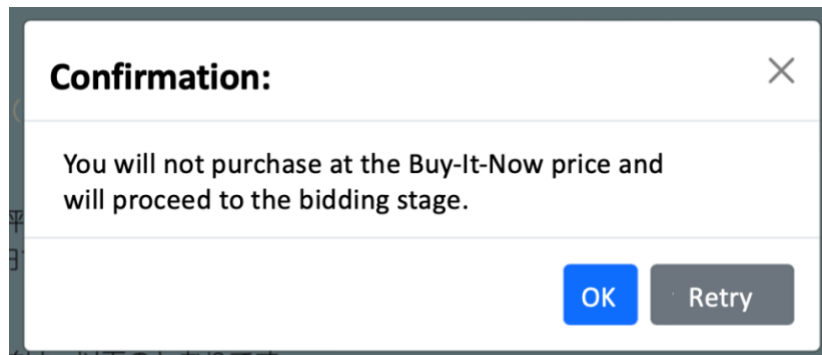
NO

- If you wish to win the auction at the Buy-It-Now price, click “YES.” If you do not wish to win the auction, click “NO.”
- After making your selection, a confirmation screen will appear. If everything is correct, click “OK.” If you wish to change your selection, click “Retry” to make your choice again.

(1) If you select “YES”



(2) If you select “NO”



- This selection will be made simultaneously by all members of your group.
- If only one person in your group selects “YES,” that person will be the winner of the auction.
- If multiple people select “YES,” one person will be randomly chosen from them with equal probability to be the winner.
- Once a winner is determined at the Buy-It-Now price, the auction for that round will end. The winner will earn a profit equal to their evaluation value minus the Buy-It-Now price.

$$\text{Profit} = \text{Evaluation Value} - \text{Buy-It-Now Price}$$

- Example: If your evaluation value is JPY 820 and you are the only person to select “YES,” you will be the winner. In this case, you will earn a profit of JPY 820 – JPY 740 = JPY 80. The other participants will receive a profit of a JPY 0.
- If even one person selects “YES” for the Buy-It-Now price, the next screen will appear after all participants have made their selections.
- On this screen, you can review the results of each participant’s decision regarding the Buy-It-Now option and see the profit you have earned in this round.

[T5, T6]

- Therefore, you will know how many people participated in the auction.

(1) If you are the Winner

Buy-It-Now Stage Results (Round 3, 4 Participants)

- You purchased the item at the Buy-It-Now price.
- Your profit for this round is as follows:
Evaluation Value – Buy-It-Now Price = Profit

780 – 740 = 50 yen

Buy-It-Now Decisions by Participants:

Participant 1: **YES** (You)

Participant 2: **NO**

Participant 3: **YES**

Participant 4: **NO**

[Next](#)

(2) If you are not the Winner

Buy-It-Now Stage Results (Round 3, 4 Participants)

- You did not win the auction.
- Your profit for this round is **0 yen**.

Buy-It-Now Decisions by Participants:

Participant 1: **YES**

Participant 2: **NO** (You)

Participant 3: **YES**

Participant 4: **NO**

[Next](#)

- If no one selects “YES” for the Buy-It-Now price, the auction will proceed to the normal bidding stage.
- There are two possible scenarios when the auction moves to this stage:
 1. All group members' evaluation values will *decrease to* [T1, T3, T5] *60% of their original evaluation values* from the Buy-It-Now stage.
[T2, T4, T6] *20% of their original evaluation values* from the Buy-It-Now stage.
 2. All group members' evaluation values will *remain the same as their original evaluation values* from the Buy-It-Now stage.
[T1, T2, T5, T6]
- Which of these two scenarios will occur is determined randomly for each auction.
[T3, T4]
- However, *until the bidding stage begins, it will not be clear whether situation 1 or 2 will occur.*
- *The probabilities of each situation occurring are also unknown beforehand.*

Example: Let's consider a two-person group auction. In the Buy-It-Now stage, your evaluation value is a JPY 800 and your opponent's evaluation value is a JPY 500.

1. If the evaluation values decrease in the auction and no one selects the Buy-It-Now price, your evaluation value will decrease to

[T1, T3, T5] $\text{JPY } 800 \times 0.6 = \text{JPY } 480$, and your opponent's evaluation value will decrease to $\text{JPY } 500 \times 0.6 = \text{JPY } 300$ in the subsequent bidding stage.

[T2, T4, T6] $\text{JPY } 800 \times 0.2 = \text{JPY } 160$, and your opponent's evaluation value will decrease to $\text{JPY } 500 \times 0.2 = \text{JPY } 100$ in the subsequent bidding stage.

2. If the evaluation values remain unchanged in the auction, your evaluation value will stay at a JPY 800, and your opponent's evaluation value will remain at a JPY 500, even in the bidding stage.

[T1, T2, T5, T6]

- Whether the evaluation values change in the subsequent bidding stage can be confirmed in the Buy-It-Now stage.

[T3, T4]

- The evaluation value for each situation, 1 and 2, can be confirmed in advance on the Buy-It-Now stage screen.

(1) If the Evaluation Values Decrease

Buy-It-Now Stage (Round 3, 4 Participants)

- Your evaluation value for this item is **790 yen**.
- If you proceed to the bidding stage without using the Buy-It-Now option, **your evaluation value decreases to 60% of its current value (474 yen)**.
- The Buy-It-Now price for this item is **740 yen**.
- If you win, your profit will be calculated as follows:

Evaluation Value – Buy-It-Now Price
= 790 yen – 740 yen = 50 yen

- Will you purchase at the Buy-It-Now price?
(Time remaining: 0:22)

(2) If the Evaluation Values Remain the Same

Buy-It-Now Stage (Round 3, 4 Participants)

- Your evaluation value for this item is **790 yen**.
- If you proceed to the bidding stage without using the Buy-It-Now option, **your evaluation value will remain 790 yen**.
- The Buy-It-Now price for this item is **740 yen**.
- If you win, your profit will be calculated as follows:

Evaluation Value – Buy-It-Now Price
= 790 yen – 740 yen = 50 yen

- Will you purchase at the Buy-It-Now price?
 (Time remaining: 0:22)

- Please note that the Buy-It-Now stage has a time limit, and the remaining time will always be displayed in the top-right corner of the screen.
- If no selection is made within 90 seconds, “NO” will be automatically selected, so please be aware of this.

Bidding Stage :

- Each member of the group will select a “*Bid Amount.*” In the following screen, enter a bid amount as an integer between 0 and 1000, then press “*Confirm.*”
- The following screen displays your evaluation value in the bidding stage.

[T5, T6]

- The number of participants in the same auction group (2 or 4 people) is also specified.

Bidding Stage (Round 3, 4 Participants)

- No one purchased at the Buy-It-Now price, so the auction will proceed to the bidding stage.
- If you win the auction, you will pay the second-highest bid amount.
- Your profit if you win the auction will be as follows:

Evaluation Value (747 yen) – Second-Highest Bid

- Please decide your bid amount. (Time remaining: 0:47)

 ¥

- A confirmation screen will then appear.
- If your bid amount is correct, press “OK.” If you wish to change your bid amount, press “Rebid.”

Confirmation: ×

Is your bid amount **790 yen** correct?

- The person with the highest bid amount will be the winner of the auction.
- If multiple people select the same highest bid amount, one person will be randomly chosen from them with equal probability to be the winner.
- Once the winner is determined in the bidding stage, the winner will earn a profit equal to their evaluation value in the bidding stage minus the **“second-highest bid amount.”**

Profit = Evaluation Value in the Bidding Stage – Second-Highest Bid Amount

- If multiple people select the highest bid, that bid will be considered the “second-highest bid.”

Example: Let’s consider a two-person group auction, where your evaluation value is a JPY 700.

(1) Let’s assume **you place a bid lower than your evaluation value, for example, JPY 500**. In this case:

- *If your opponent places a bid lower than JPY 500, for example, JPY 400, you will be the winner. You will pay the second-highest bid of a JPY 400 and win the item with an evaluation value of a JPY 700, so your profit will be $\text{JPY } 700 - \text{JPY } 400 = \text{JPY } 300$.*
- *If your opponent places a bid higher than **JPY 500**, for example, JPY 600, the opponent with the highest bid will be the winner. Since you did not win the auction, your profit will be a JPY 0.*

(2) Let’s assume **you place a bid equal to your evaluation value, for example, JPY 700**. In this case:

- *If your opponent places a bid lower than **JPY 700**, for example, JPY 600, you will be the winner. You will pay the second-highest bid of JPY 600 and win the item with an evaluation value of JPY 700, so your profit will be $\text{JPY } 700 - \text{JPY } 600 = \text{JPY } 100$.*
- *If your opponent places a bid higher than **JPY 700**, for example, JPY 800, the opponent with the highest bid will be the winner. Since you did not win the auction, your profit will be a JPY 0.*

Remark: When comparing scenarios (1) and (2), please note that by placing a bid equal to your evaluation value (JPY 700) rather than a lower bid (e.g., JPY 400), your chances of winning the auction increase.

(3) Let’s assume **you place a bid higher than your evaluation value, for example, JPY 900**. In this case:

- *If your opponent places a bid lower than **JPY 900**, for example, JPY 800, you will be the winner. However, you will pay the second-highest bid of JPY 800 and win the item with an evaluation value of JPY 700. As a result, your profit will be $\text{JPY } 700 - \text{JPY } 800 = -\text{JPY } 100$, which means you incur a loss.*

- If your opponent places a bid higher than **JPY 900**, for example, JPY 1000, the opponent with the highest bid will be the winner. Since you did not win the auction, your profit will be a JPY 0.

Remark: As you can see, if you place a bid higher than your evaluation value (e.g., JPY 900), it's possible that your profit will be negative.

- Based on the above, in the bidding stage, *placing a bid* **EQUAL TO YOUR EVALUATION VALUE** increases your chances of winning the auction compared to bidding below your evaluation value. Additionally, by not bidding above your evaluation value, you avoid the risk of incurring a negative profit. Therefore, bidding the amount equal to your evaluation value is the **best choice**.
- After all participants have placed their bids, the next screen will display the bidding results along with the profit you will earn from this auction.

(1) If you are the Winner

Buy-It-Now Stage Results (Round 3, 4 Participants)

- You won this auction.
- Your profit for this round is as follows:
Evaluation Value – Second-Highest Bid = Profit

474 – 462 = 12 yen

The bid amounts of each participant:

Participant 1: **504** (You) ← Winner

Participant 2: **462**

Participant 3: **192**

Participant 4: **120**

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(2) If you are not the Winner

Bidding Stage Results (Round 3, 4 Participants)

- You did not win the auction.
- Your profit for this round is **0 yen**.

The bid amounts of each participant:

Participant 1: **504** ← Winner
Participant 2: **462** (You)
Participant 3: **192**
Participant 4: **120**

Next

- Please note that the bidding stage has a time limit, and the remaining time will always be displayed in the top right corner of the screen.
- If the bidding is not completed within 60 seconds, the amount entered at that point (or JPY 0 if left blank) will automatically be submitted as your bid, so please be aware of this.

Summary of the Experiment Procedure:

- Each auction will proceed in the following steps:
 1. You will be informed of the group size (either 2 or 4 people).
 2. The Buy-It-Now price (JPY 740) and your evaluation value will be displayed.
 3. If you wish to win at the Buy-It-Now price, select “YES;” if not, select “NO.”
 4. If at least one person in the group selects “YES,” a winner will be randomly chosen from those who selected “YES.” The winner will pay the Buy-It-Now price, and the auction will end. If no one selects “YES,” the process moves to step 5.
[T3, T4] *However, at this point, it is unclear whether the evaluation value at the bidding stage will remain as shown in step 2 or if it will decrease to 60% of that value, as the probability of either scenario occurring is unknown.*
 5. [T1, T2] You will select your bid amount.
[T3, T4] After the evaluation value in step 5 is revealed, you will then select your

bid amount.

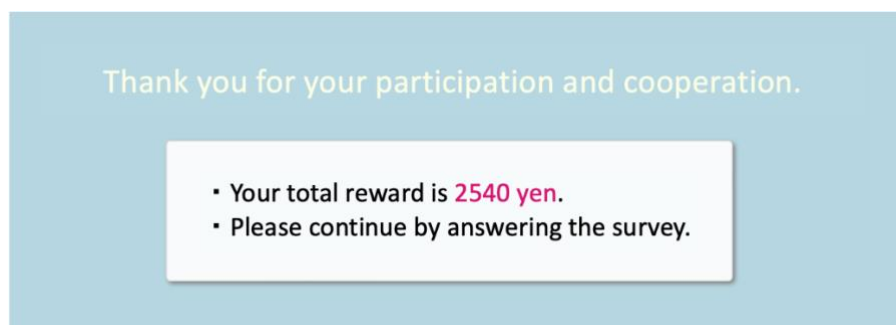
[T5, T6] After the number of participants in the auction is revealed, you will then select your bid amount.

[T1, T3, T5] (in some auctions, your evaluation value will decrease to 60 % of the Buy-It-Now price).

[T2, T4, T6] (in some auctions, your evaluation value will decrease to 20 % of the Buy-It-Now price).

6. The person with the highest bid will win and pay the second-highest bid amount.

- You will repeat these auctions multiple times, with changes in the evaluation value, as well as the number of participants and group members each time.
- Once all the auctions are finished, the next screen will appear. This screen will show the cumulative amount of profit you earned from each auction.



- This total amount will be the reward you will receive for today's experiment.
- After the auction experiment is completed, while we are tallying your reward amount, you will be asked to complete a survey.
- If you answer all the questions, an additional JPY 1000 will be added to the reward displayed on the confirmation screen above.
- The total reward will be paid to you via Amazon gift card.

Note:

- Please keep connected to Zoom throughout the experiment.
- During the experiment, do not consult or discuss with others.
- This experiment is not a test, so there are no right or wrong answers. Please make your selections based on your own judgment.
- Do not use any calculation aids such as calculators or notes when making your choices.